

EXP 10

FIFO & LRU Page Replacement

AIM:

To find out the number of page faults that occur using First-in First-out (FIFO) page replacement technique and a c program to implement LRU page replacement algorithm.

1. FIFO Page Replacement

ALGORITHM

1. Start
2. Get number of pages
3. Get page reference string
4. Get number of frames
5. Initialize all frames as empty
6. Check each page one by one
7. If page is not in frame, page fault occurs
8. Replace the oldest page using FIFO order
9. Display frame contents after each step
10. Display total page faults
11. End

CODE:

```
#include <stdio.h>
```

```
int main() {  
    int pages[50], frames[10];  
    int n, f, i, j, k = 0;  
    int pageFaults = 0, found;  
  
    printf("Enter number of pages: ");  
    scanf("%d", &n);  
  
    printf("Enter page reference string:\n");  
    for(i = 0; i < n; i++) {
```

```

    scanf("%d", &pages[i]);
}

printf("Enter number of frames: ");
scanf("%d", &f);

for(i = 0; i < f; i++) {
    frames[i] = -1;
}

printf("\nPage\tFrames\n");

for(i = 0; i < n; i++) {
    found = 0;

    for(j = 0; j < f; j++) {
        if(frames[j] == pages[i]) {
            found = 1;
            break;
        }
    }

    if(found == 0) {
        frames[k] = pages[i];
        k = (k + 1) % f;
        pageFaults++;
    }

    printf("%d\t", pages[i]);

    for(j = 0; j < f; j++) {
        if(frames[j] != -1)
            printf("%d ", frames[j]);
        else
            printf("- ");
    }

```

```

        printf("\n");
    }

    printf("\nTotal Page Faults = %d\n", pageFaults);

    return 0;
}

```

2. LRU Page Replacement

ALGORITHM

1. Start
2. Get number of pages
3. Get page reference string
4. Get number of frames
5. Initialize all frames as empty
6. Check each page one by one
7. If page is already present, update its recent usage
8. If page is not present, page fault occurs
9. Replace the least recently used page
10. Display frame contents after each step
11. Display total page faults
12. End

CODE:

```

#include <stdio.h>

int main() {
    int pages[50], frames[10], recent[10];
    int n, f, i, j, pos;
    int pageFaults = 0, found, min;

    printf("Enter number of pages: ");
}

```

```
scanf("%d", &n);
```

```
printf("Enter page reference string:\n");  
for(i = 0; i < n; i++) {  
    scanf("%d", &pages[i]);  
}
```

```
printf("Enter number of frames: ");  
scanf("%d", &f);
```

```
for(i = 0; i < f; i++) {  
    frames[i] = -1;  
    recent[i] = 0;  
}
```

```
printf("\nPage\tFrames\n");
```

```
for(i = 0; i < n; i++) {  
    found = 0;
```

```
    for(j = 0; j < f; j++) {  
        if(frames[j] == pages[i]) {  
            found = 1;  
            recent[j] = i + 1;  
            break;  
        }  
    }  
}
```

```
if(found == 0) {  
    pageFaults++;
```

```
    pos = -1;
```

```
    for(j = 0; j < f; j++) {  
        if(frames[j] == -1) {  
            pos = j;
```

```

        break;
    }
}

if(pos == -1) {
    min = recent[0];
    pos = 0;

    for(j = 1; j < f; j++) {
        if(recent[j] < min) {
            min = recent[j];
            pos = j;
        }
    }
}

frames[pos] = pages[i];
recent[pos] = i + 1;
}

printf("%d\t", pages[i]);

for(j = 0; j < f; j++) {
    if(frames[j] != -1)
        printf("%d ", frames[j]);
    else
        printf("- ");
}

printf("\n");
}

printf("\nTotal Page Faults = %d\n", pageFaults);

return 0;
}

```

RESULT:

Thus, FIFO and LRU page replacement algorithms were successfully implemented and executed.